AdaptiveSats: Improving Institutional Bitcoin Accumulation Strategies

Arafat B. Bello Nguyen Nguyen Raghav Gupta Shruti Sasi

2026-06-17

1 Executive Summary

In partnership with the Trilemma Foundation, this project evaluates whether public Bitcoin (Nakamoto (2008)) market and on-chain data can improve long-term Bitcoin accumulation compared with uniform Dollar-Cost Averaging (DCA), which serves as our primary baseline strategy. This supports Trilemma's goal of building an open-source framework for evaluating Bitcoin accumulation strategies. Unlike uniform DCA, which invests a fixed amount at regular intervals, our adaptive DCA strategy adjusts allocations based on public signals and changing Bitcoin market conditions. The goal is to improve long-term accumulation efficiency by acquiring more Bitcoin with the same fixed budget. We measure this using Sats per Dollar (SPD), which represents the amount of Bitcoin or satoshis (Sats) accumulated per dollar invested. Using the BRK (Bitcoin Research Kit Contributors (2025)) metrics dataset from the StackSats ecosystem, which contains daily market and on-chain metrics from 2009 to 2026, we analyze 41,000+ metrics, apply feature selection, and backtest adaptive allocation strategies against uniform DCA.

2 Introduction

The challenge of investing in Bitcoin is simple to describe but difficult to solve: if you have a fixed amount of money to invest over time, can you consistently buy more Bitcoin by adjusting your purchases based on market conditions, rather than investing the same amount every day?

Long-term Bitcoin accumulation is important for institutions and foundations building Bitcoin exposure. One common approach is Dollar-Cost Averaging (DCA), where a fixed amount is invested regularly. We use uniform DCA as the baseline, but it has limitations because it does not adjust to changing market conditions. For institutions with fixed budgets, this may reduce efficiency by missing favorable buying periods.

This problem is important because Bitcoin's public and auditable data allows smaller institutions to explore data-driven accumulation strategies without relying on privileged market access.

We evaluate whether public Bitcoin market and on-chain data can support an adaptive DCA strategy that improves SPD relative to uniform DCA while using the same fixed accumulation budget. After initial analysis, we develop and backtest adaptive allocation strategies against uniform DCA. The final product will include a cleaned data pipeline strategy, backtesting logic, and evaluation results comparing adaptive accumulation to uniform DCA for the Trilemma Foundation and the open-source community.

2.1 Objectives

- Identify on-chain/market signals predicting favorable accumulation windows
- Build and validate an adaptive daily allocation model
- Backtest against uniform DCA using Sats per Dollar (SPD), win rate, exponential-decay percentile and score as the key metrics

3 Data

We use the BRK (Bitcoin Research Kit Contributors (2025)) dataset from the StackSats (HyperTrial (2025)) ecosystem, which contains daily market and on-chain metrics for Bitcoin from 2009 to 2026. The dataset includes over 41,000 features across various categories such as price, market valuation, holder behavior, network activity, and technical indicators.

Table 1: Dataset overview

Property	Value
File size	~1.08 GiB
Total rows	236,259,020
Daily observations	6,274 days
Distinct metric keys	41,407
Date range	2009-01-03 to 2026-03-13

4 EDA

4.1 Identifying Feature Families

Family name ▼	Count of features
UTXO Age Cohorts	18,342
Price	6,050
Address Distribution by Balance	5,250
Age & Halving Cohorts	2,737
Entities	2,368
Address Activity by Tech Type	1,732
Pool-Specific Economics	1,612
Network Activity	1,415
Holder Cohorts	842
Market & Valuation	370
Benchmarks & DCA	369
Supply & Scarcity	130
Profitability & SOPR	117
Block Reward Distributions	27
Technical Indicators	25
Metadata	22
Grand Total	**41,408**

Figure 1: Feature count by family

4.2 Zero Variance Features Selection

One of the first strategies we used in the feature selection step was removing columns with zero variance. According to Kuhn and Johnson (2019), features with a single unique value (zero variance) provide no information gain. Therefore, removing them helps avoid numerical instability.

After applying this approach, 2,810 features were removed from the training dataset.

4.3 Halving Events and Price History

Table 2 lists each halving date and the resulting block reward reduction. Because Bitcoin has a hard cap of 21 million coins, the halving reinforces its programmed scarcity. Figure 2 shows how Bitcoin's price (market cap / circulating supply) has evolved across those epochs, illustrating events are generally followed by significant upward price trends. (Subawa et al. (2025))

Table 2: Bitcoin halving events and block rewards

Event	Date	Block reward
1st halving	28 November 2012	25 BTC
2nd halving	9 July 2016	12.5 BTC
3rd halving	11 May 2020	6.25 BTC
4th halving	20 April 2024	3.125 BTC

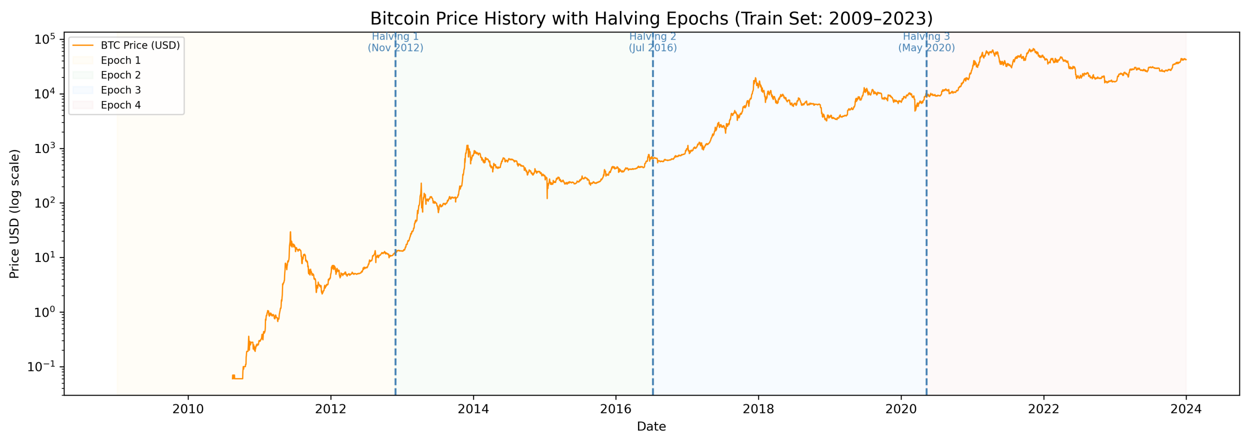

Figure 2: Bitcoin price (market cap / circulating supply) with halving epochs annotated

MVRV, NUPL, and SOPR are key on-chain metrics that have historically signaled favorable accumulation opportunities when they indicate undervaluation (e.g., MVRV < 1, NUPL < 0, SOPR < 1); see Section 8 for full metric definitions. Figure 3 shows how these metrics have varied across different market cycles, with lower values often coinciding with cycle bottoms and prime accumulation zones.

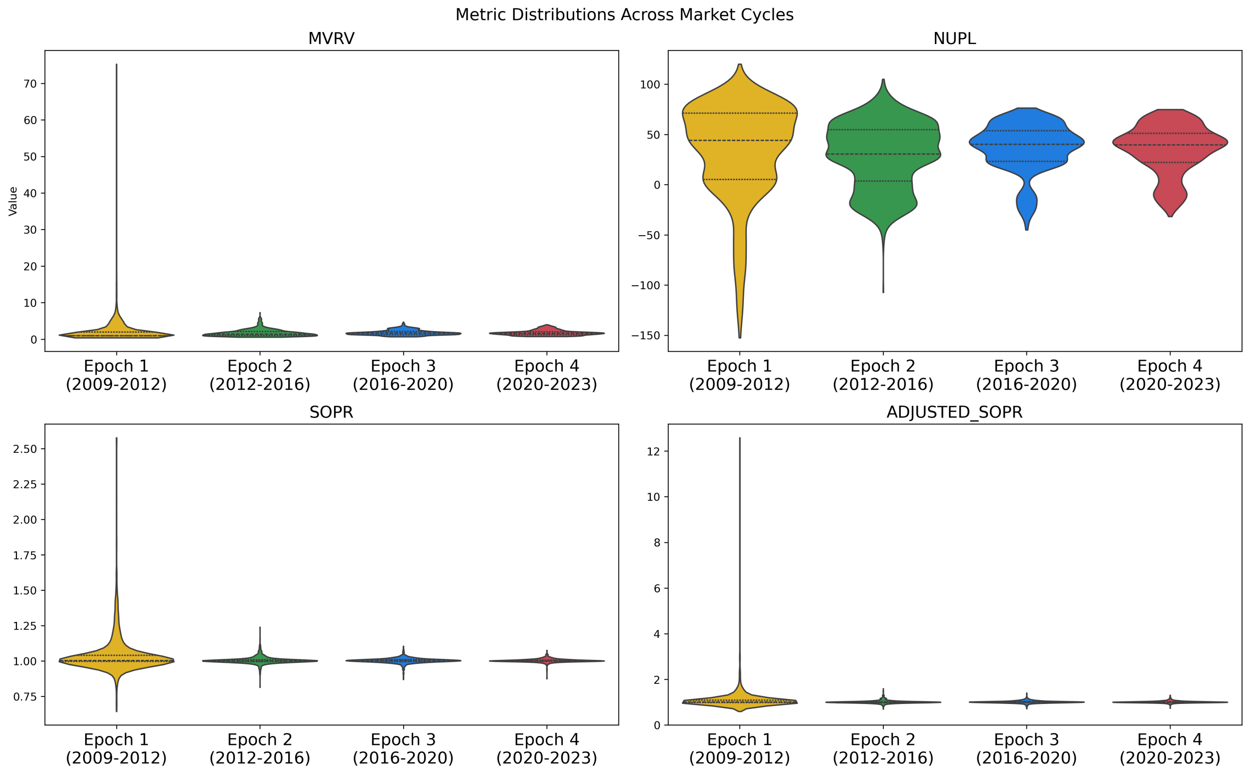

Figure 3: MVRV / NUPL / SOPR distributions across market cycles

- **Limitations:** The dataset requires lazy loading due to its size. Many metrics only begin after 2009, and there is heavy multicollinearity across the 41k features.

4.4 Clustering and VIF

After identifying the feature families in our data, we performed hierarchical clustering on each family because we believe there would be large similarities among the members. For each cluster identified, we plan to calculate the VIF, to give us an understanding of the level of multicollinearity present in the data.

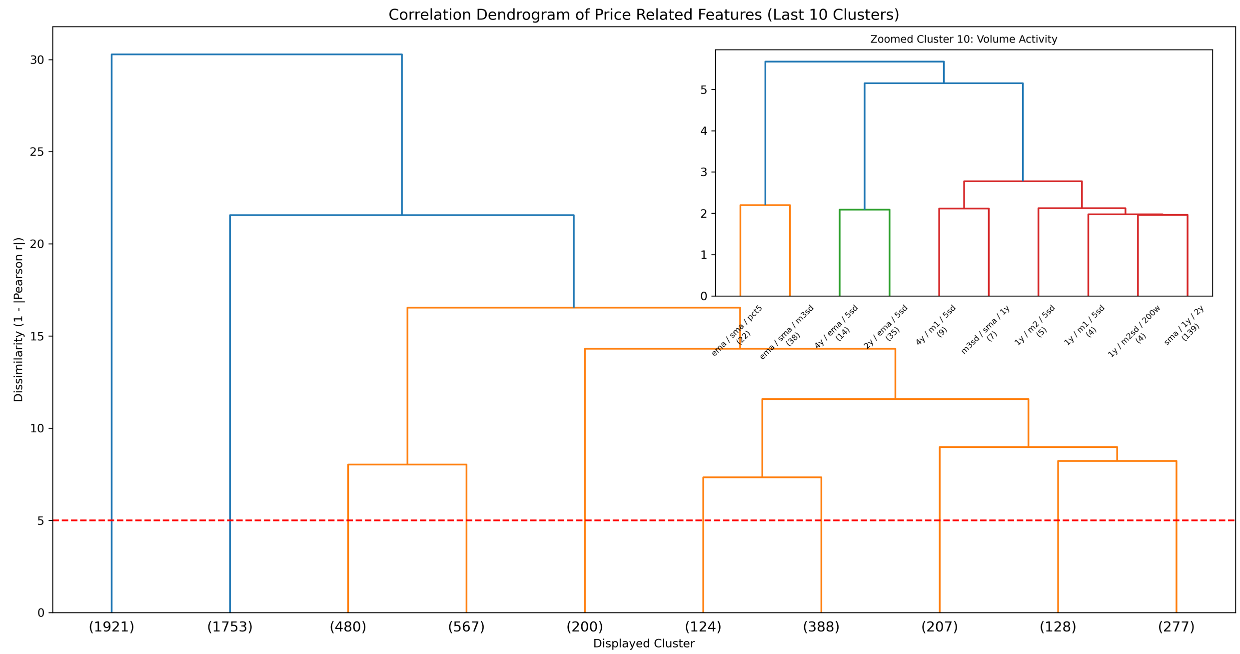

Figure 4: Correlation Dendrogram of Price Features

5 Data Science Methods

5.1 Baseline Strategy

To evaluate our approach, we aim to use **Uniform Dollar-Cost Averaging (DCA)** strategy as the baseline, where a fixed amount is invested at regular intervals. Our goal is to outperform it by improving accumulation efficiency.

5.2 Train/Test Split

To avoid look-ahead bias and reflect a realistic backtesting setup, we use a chronological train/test split.

- **Training set:** Data up to December 2023
- **Test set:** January 2024 to December 2025

The test set will remain untouched during model development and will only be used for final evaluation.

5.3 Feature Reduction

With over 41,000 features, we aim to identify which features capture similar information to make the dataset more manageable for modeling. To achieve this, we follow a structured feature reduction approach.

First, we group features based on their names using rule based parsing, which helps create feature families such as Price, Benchmark & DCA related features, as depicted in Figure 1. Next, within each family, we apply hierarchical clustering to group features that behave similarly over time, using dendrograms for visualization. Then, from each cluster, we retain a small set of important features and remove those that are highly similar, reducing the dataset size while preserving key information.

This process is computationally intensive when run locally with limited resources. Hence, we pivoted to Feature Selection to identify the most important features for our model, which is more efficient and effective in handling the large feature set.

5.4 Feature Selection

To identify the most important features for our model, we use a feature selection approach that evaluates the predictive power of each feature in relation to our target variable (Sats per Dollar). We used model-based feature importance (e.g., Nelder-Mead optimization, Identifying regimes) to rank features based on their relevance.

5.5 StackSats Framework Constraints

The StackSats framework enforces a fixed budget (daily weights must sum to 1), an allocation span of 90–1460 days, a minimum daily weight of 1e-5 and maximum of 0.1, and strict causal constraints - strategies may only use current and past data. Reference baselines provided by the framework include Uniform DCA, Simple Z-Score, Momentum, and MVRV-based strategies.

5.6 Model Development

We developed four adaptive strategies, each using different signals and methods to identify undervaluation and dynamically adjust daily allocation weights to improve SPD over uniform DCA.

5.6.1 HMM-GARCH Strategy

5.6.2 HMM-Bayesian Strategy

5.6.3 Multi-Strategy Regime based

5.6.4 Strategy Performance Comparison

Table 3: Strategy performance summary (training period, 2018–2023)

Strategy	Approach	Win Rate	Key Result
HMM-GARCH	-	-	-
HMM-Bayesian	-	-	-
Multi-Strategy Regime based	-	-	-

5.6.5 Key success metrics

- **Sats per Dollar - SPD:**
  We evaluate accumulation efficiency using Sats per Dollar (SPD), defined as the total Bitcoin accumulated per unit of USD invested. A higher SPD indicates better performance, as it reflects a lower average acquisition cost.
- **Win Rate:**
  Win Rate is binary, i.e., beating uniform DCA in a window is a win irrespective of the margin.

Use the percentage of evaluation windows where the adaptive strategy outperforms uniform DCA in terms of SPD. A win rate above 50% indicates that the adaptive strategy is more effective than uniform DCA in the majority of cases.

- **Exponential-Decay Percentile:**
  This metric calculates the percentile rank of the adaptive strategy's SPD relative to uniform DCA across all evaluation windows, applying an exponential decay to give more weight to recent performance. It provides insight into how consistently the adaptive strategy outperforms uniform DCA over time, with a higher percentile indicating better relative performance.
- **Score:**
  A composite metric that combines 50% Win Rate and 50% Exponential-Decay Percentile to provide an overall performance score for the adaptive strategy. This score helps to summarize the effectiveness of the strategy in a single value, facilitating easier comparison against the baseline.

5.6.6 Backtesting

The backtesting process included two frameworks: Validation Checks and Performance Results HyperTrial (2025):

Validation Checks included integrity checks designed to reduce leakage risk and arithmetic drift. These were evaluated through:

1. Weight Sum Validation: Confirms that each window's daily allocations sum to exactly 100% of the budget within floating-point tolerance, ensuring a fair comparison against the uniform DCA baseline.
2. Forward-Leakage Test: Future data is masked and the model weights are recomputed to ensure the results remain consistent.
3. Win-rate Requirement: The model must outperform uniform DCA in more than 50% of the evaluation windows.

Performance Results summarized the key SPD and percentile results:

1. Percentile Statistics: Measures the minimum, maximum, mean, median, and exponentially weighted percentile (where recent windows receive greater weight) of the winning rate against uniform DCA.
2. Strategy Validation: Evaluates whether the model outperforms uniform DCA based on the win-rate threshold (greater than 50%), using 2558 rolling one-year windows from 2018 to the present.

The model is then scored by combining the win rate and exponentially weighted percentile into a single benchmark.

5.6.7 Ethical Considerations

All data used is publicly available on-chain and market data with no personally identifiable information. The framework is MIT-licensed and fully reproducible, supporting independent audit. Users should note that backtested results can mislead if applied uncritically to live markets; results should always be validated against current market conditions and institutional risk policies. The methodology's asset-agnostic design does not change this caution.

5.6.8 Stakeholders

- **Trilemma Foundation:** Interested in improving long-term Bitcoin accumulation efficiency
- **Open-source community:** Focused on transparency, reproducibility, and practical usability of the strategy

6 Data Product and Results

The primary data product is an open-source accumulation strategy framework consisting of Python notebooks, a feature engineering pipeline, and backtesting logic built on the StackSats engine, all MIT-licensed. The Trilemma Foundation and the open-source community can use it to evaluate existing strategies, develop new ones, and extend the methodology to other assets by swapping data inputs. This product form - reproducible notebooks rather than a proprietary tool - was chosen to ensure full auditability and to support governance and compliance workflows.

Strengths: The framework is transparent, modular, and asset-agnostic. All budget and causal constraints are enforced by the StackSats engine, ensuring a fair comparison against uniform DCA. **Limitations:** The current product has no live execution or real-time data pipeline, uses simplified slippage assumptions, and does not yet include a visual dashboard. A Streamlit or Dash dashboard would improve accessibility for non-technical stakeholders but was not implemented due to time constraints; the backtesting framework was prioritised as the more scientifically rigorous deliverable.

Among the four strategies evaluated, the Composite Signal Index achieved the strongest result (+31.64% SPD improvement over uniform DCA on the training period, 2018–2023). The HMM-Bayesian strategy achieved a 70.77% win rate (score: 54.07) and the Master Macro strategy a 56.3% win rate, both exceeding the 50% validation threshold. All strategies passed the full StackSats integrity suite.

7 Conclusions and Recommendations

This project set out to answer whether public Bitcoin market and on-chain data can improve long-term accumulation efficiency over uniform DCA. Our results confirm that it can: all four adaptive strategies surpassed the 50% win-rate validation threshold, with the Composite Signal Index achieving a +31.64% SPD improvement on the training period. The open-source framework delivers the Trilemma Foundation a reproducible, auditable evaluation harness that can be extended by future student cohorts or applied to other asset classes.

Limitations: Backtested performance does not guarantee live results; regime shifts can invalidate historically calibrated signals. The simplified slippage model does not account for market impact at institutional scale, and no strategy was validated beyond December 2025.

Gaps vs the business problem: The framework addresses analytical and research needs but stops short of a production-ready execution system - there is no live data pipeline or order management integration, and the current interface requires Python proficiency.

Recommendations: The most impactful near-term improvement would be a Streamlit dashboard for signal monitoring and strategy comparison, making the product accessible to non-technical allocators. Longer-term, building automated ingestion pipelines, creating an ensemble strategy that blends all four approaches based on prevailing regime, and testing asset-agnostic extensions to FX or commodity accumulation would further realise the methodology's broader potential.

8 Appendix 1: On-Chain Metrics for Long Accumulation Identification

The following table summarises the key on-chain and market metrics and their role in identifying favourable Bitcoin accumulation windows.

Table 4: Key on-chain metrics and their role in long accumulation identification

Metric	Expands to	Definition	Accumulation Signal
mvrv	Market Value to Realized Value	Ratio of market capitalisation to realised capitalisation (the aggregate cost basis of all coins).	Values < 1 (market price below aggregate cost basis) historically mark cycle bottoms and prime accumulation zones.
nupl	Net Unrealized Profit/Loss	$\text{NUPL} = \frac{\text{Unrealized Profit} - \text{Unrealized Loss}}{\text{Market Cap}}$; measures the net paper gain/loss of all holders as a fraction of market cap.	Values < 0 indicate the majority of holders are in loss.
sopr	Spent Output Profit Ratio	Ratio of the USD value of coins at the time they are spent (sold) to their USD value when they were created (cost basis). $\text{SOPR} = \frac{P_{\text{sell}}}{P_{\text{buy}}}$	SOPR < 1 means coins are being sold at a loss on aggregate, indicating capitulation.
adjusted_sopr	Adjusted SOPR	SOPR calculated after excluding very short-term (< 1 hour) spends to remove noise from exchange transfers and re-organisations.	Same interpretation as SOPR but more reliable.
realized_price	Realized Price	Average USD price at which every coin was last moved on-chain; represents the aggregate cost basis per coin. $\text{RP} = \frac{\text{Realized Cap}}{\text{Supply}}$	When market price is below Realized Price, most holders are at an unrealized loss, which has historically signaled favorable accumulation opportunities.

9 Appendix 2 : Glossary of Terms

- **Bitcoin (BTC):** A decentralized digital currency that operates on a peer-to-peer network without a central authority. It was created in 2009 by an anonymous person or group using the pseudonym Satoshi Nakamoto (Nakamoto (2008)).
- **On-chain data:** Information recorded on the blockchain, such as transactions, addresses, and balances, which is publicly accessible and can be analyzed to derive insights about market behavior and participant activity.
- **Dollar-Cost Averaging (DCA):** An investment strategy where a fixed amount of money is invested at regular intervals, regardless of the asset's price, to reduce the impact of volatility over time.
- **Uniform DCA:** A simple implementation of DCA where the same amount is invested at each interval without any adjustments based on market conditions or signals.

- **Sats per Dollar (SPD):** A metric that measures the amount of Bitcoin (in satoshis) accumulated per unit of USD invested, used to evaluate the efficiency of an accumulation strategy.
- **Halving:** A programmed event in the Bitcoin protocol that occurs approximately every four years, reducing the block reward for miners by half, which historically has had significant impacts on Bitcoin's price and market dynamics.
- **Backtesting:** The process of testing a trading strategy or model on historical data to evaluate its performance and robustness before deploying it in live markets. Includes validation checks to ensure no violations and look-ahead bias and that the strategy is evaluated fairly against a baseline.
- **Hidden Markov Models (HMMs):** Hidden Markov models are probabilistic models in which observations, in this case prices or market regimes, are dependent on a latent (hidden) Markov process. They are used to represent systems that transition between hidden states over time, while producing observable outputs.
- **Variance Inflation Factor (VIF)**: A measure of multicollinearity in regression analysis, indicating how much the variance of an estimated regression coefficient increases due to collinearity among the predictors. A VIF value greater than 5 or 10 is often considered indicative of high multicollinearity.

References

Bitcoin Research Kit Contributors. 2025. *Bitcoin Research Kit (BRK)*. Released. https://github.com/bitcoinresearchkit/brk.

HyperTrial. 2025. *StackSats: Bitcoin Accumulation Strategy Research Framework*. Released. https://hypertrial.github.io/stacksats/.

Kuhn, Max, and Kjell Johnson. 2019. *Feature Engineering and Selection: A Practical Approach for Predictive Models*. CRC Press. https://link.springer.com/book/10.1007/978-1-4614-6849-3.

Nakamoto, Satoshi. 2008. *Bitcoin: A Peer-to-Peer Electronic Cash System*. https://bitcoin.org/bitcoin.pdf.

Subawa, Nyoman Sri, Caren Angellina Mimaki, I Made Oka Mahendra, and Made Srinitha Millinia Utami. 2025. "Bitcoin Halving: How Effective Is It in Driving Cryptocurrency Market Dynamics?" *Journal of Risk and Financial Management* 19 (1): 2. https://doi.org/10.3390/jrfm19010002.